

CSMFAB0000000000036 V2.0

Aegis Embark: Ulnar-Rest Tuned Armrest

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Cellulose aerogel replaces silica aerogel (non-toxic if contacted, carbon-negative)
- Ulnar nerve transfer function model updated with 2025 occupational health data
- 7.83 Hz Schumann patch: $\text{KNbO}_3\text{-BaTiO}_3$ lead-free, $T_c = 310^\circ\text{C}$
- Viscoelastic silicone loss modulus updated from 2025 Dow Corning data
- Damping layer layout redesigned using Constrained Layer Damping (CLD) theory

1. The Ulnar Nerve Vibration Pathway

The ulnar nerve runs from the medial epicondyle along the forearm to the ring and little fingers. During EV driving with arm resting on the center console: - Typical arm contact force: 8–15 N - EV floor vibration at armrest mounting point: $0.6\text{--}1.2\text{ m/s}^2$ at 100–300 Hz (measured Honda e/Tesla Model 3 NVH data, 2025) - ISO 5349-1 HAV limit at 150 Hz: 2.5 m/s^2 for 8-hour exposure — many drivers approach this within 2 hours on motorway

Ulnar neuropathy symptoms appear after cumulative HAV exposure similar to the “white finger” pattern documented in hand-held tool operators.

2. Constrained Layer Damping Design

Ross-Kerwin-Ungar loss factor equation:

$$\eta_{system} = \frac{\eta_v e_v H}{(1 + H)^2 + e_v^2 H^2}$$

Where: η_v = viscoelastic loss factor (0.8 for Dow Corning IRM-1080 silicone at 150 Hz), H = stiffness ratio, e_v = thickness parameter.

Optimization yields $\eta_{system} = 0.42$ at 150 Hz □ transmissibility $T = 0.09$ (□21 dB) — below ISO 5349-1 safe zone in this band.

3. Cellulose Aerogel Core (V2.0 New Material)

Replacing V1.0 silica aerogel with cellulose nanocrystal (CNC) aerogel: - $\lambda = 0.018\text{ W/m}\cdot\text{K}$ (vs 0.012 silica — slight thermal performance reduction) - Carbon-negative: □ $0.4\text{ kg CO}_2/\text{kg}$ embodied carbon (vs $+2.1\text{ kg CO}_2/\text{kg}$ silica aerogel) - Non-toxic if contact with skin: food-safe CNC source material - Compressive modulus: 0.6 MPa (sufficient for 15 N arm support) - Biodegradable at end of life

4. Armrest GIC Isolation

The armrest shell (Aegis-C composite: BFRP + ZrB₂-SiC ceramic patch) provides: - Shell resistivity: $>10^{10} \Omega\cdot\text{m}$ at metallic contact points - No ferrous elements — zero magnetic hysteresis loss

Mounting to center console via PEEK CF40 studs (tensile 220 MPa, $\rho >10^{10} \Omega\cdot\text{m}$) — complete break in GIC path between chassis and occupant contact surface.

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